



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Week 2: Git & Linux

TURTLE Incubator: SO-ARM Software Stack

Attendance



Git



What is Git

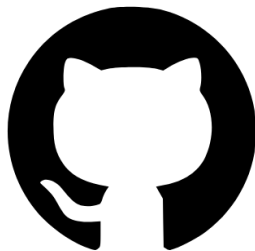


Git is the most widely used Version Control System worldwide.

Version Control Systems are incredibly useful when working on larger software projects. They allow you to:

- Work on new updates in a safe sandbox
- Easily revert changes
- Easily merge multiple contributors' updates together

What is GitHub



GitHub is a web platform that combines Git with useful collaboration tools such as:

- Pull Requests
- Code Reviews
- To-Do Lists
- Security Tools

It's kinda like a souped-up Google Drive for your Git repositories

What is Git



Git tracks any changes made to your code over time, so you can always revert back. It is a version control, meaning it is a system that tracks and manages changes to files over time.

Repository

- Where the files, folders, and change history of a project is stored.

Branches

- Your repository will contain branches, which allow you to work an experiment with code separately from the main branch. This allows you to experiment safely and catch bugs. You can use commands in git to switch between new branches and the main branch.

Push

- This will upload the code you are working on your computer to the remote repository, which will allow everyone to see your code. This is the collaborative part of Git.

Pull

- The opposite of push. This will take code from the remote repository and merge it with your local branch. This helps you keep the code you are working on in sync with other people working on the same branch.

Git Commands



The three commands you'll use every session:

- `git add .` # stage all changed files
- `git commit -m "msg"` # save a snapshot with a description
- `git push` # upload your commits to GitHub

Branches let you experiment safely:

- `git checkout -b my-feature` # new branch
- `git checkout main` # switch back to main

You can also use Git to collaborate with others by using a pull request.

Demonstration



Linux

What is Linux?



Linux is the most used operating system in the world.

While it is not used much for personal computers, it is widely used in:

- Servers
- Cloud Infrastructure
- Mobile Phones
- Supercomputers, etc.

In many scenarios, Linux is preferred over Windows for software development, as it is very easy to:

- Automate tasks
- Manage developer tools/libraries with your Package Manager

Linux also gives you virtually complete control over the operating system

What is a Distro?



Linux is technically not an OS on its own. Linux is just the **kernel**, a core part of an OS.

A **Distribution** (often shortened to **Distro**) takes Linux and combines it with everything else needed to make a full OS.

- Think of Linux as the engine and the Distro as the finished car.

When installing Linux, you first have to choose the **Distro** you want to use. Some popular options are:

- **Ubuntu <- This is what you'll probably end up using**
- Pop!_OS (if you need NVIDIA drivers)
- Fedora

Access Methods



Windows vs. Native

You can access linux through a built in software with Windows, or you can download it directly to your computer, which means it is native. Windows sacrifices some performance because it is not talking directly to the computer, but it also does not force you to reboot to switch between operating systems.

Windows Subsystem for Linux (WSL)

- Linux running as an app in Windows
- Windows
- Can be used as an extension, like Ubuntu
- Will be our main method of accessing linux

Dual Boot

- Choose between Windows or Linux to boot at startup
- Both Windows and Native
- Must reboot everytime you switch between OS
- Useful for machine learning projects for maximum GPU performance

Docker

- Runs environments called containers instead of full virtual machines
- Runs on WSL2 (Windows), not truly native
- Docker Desktop for Windows runs its containers through WSL2 under the hood, which is why WSL needs to be set up before Docker will work. Useful for cloud development or in a reproducible environment a team shares

Linux Commands



When using Linux, you'll likely need to use the terminal often.

While there are many useful commands, the most common ones you'll use are:

- `cd` (change directory)
 - `cd <path-to-directory>`
- `ls` (list all files in directory)
 - `ls <path-to-directory>`
 - You can omit the path to just list the files in your current folder
- `mv` (move a file) and `cp` (copy a file)
 - `<mv or cp> <path-to-file> <path-to-destination-directory>`

Package Managers



To install apps/tools on Linux, you'll usually use your **Package Manager**

The process will vary based on your distro, on Ubuntu this will typically look like:

- **sudo apt install** <package-name>
 - **sudo** is a keyword to execute this command with admin privileges
 - **apt** is just the name of the package manager Ubuntu uses

You'll also want to frequently update all your packages. To do this, just do:

- **sudo apt update**

Followed by:

- **sudo apt upgrade**

Demonstration



Hands On

Hands-On: Install WSL



Goal: Install WSL on your machine.

1. Open PowerShell as Administrator and run: `wsl --install -d Ubuntu`
 - a. [This one command installs WSL2, enables the Windows features it needs, and sets up Ubuntu, all at once](#)
2. Restart if prompted. Ubuntu will open on first launch and ask you to create a Unix username and password, then in that shell type:

```
sudo apt update
```

Type 'y' when it asks if you want to continue. If this window appears to hang with "Processing triggers", push enter.

3. Verify you installed gcc by executing

```
gcc --version
```

The output should be similar to this::

```
gcc (Ubuntu 11.4.0-1ubuntu1~22.04) 11.4.0
```

```
Copyright (C) 2021 Free Software Foundation, Inc.
```

```
This is free software; see the source for copying conditions. There is NO warranty; not even for  
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```